Evaluating the Effectiveness of a Behaviour-Aware Multilingual Voice-Based Local Service Discovery System Using a Multi-Agent AI Framework

Project ID
R26 -IT- 109



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Introduction (Overall Problem)

Real World Problem

- Finding reliable local services during urgent situations is often difficult in Sri Lanka. Most service discovery platforms require users to type queries in formal English and provide complete information. However, real-world communication patterns often include mixed languages such as Sinhala-English or Tamil-English and incomplete speech.

Users

- People looking for emergency services
- Vehicle repair seekers
- Home maintenance customers
- Local service providers

- **Stakeholders**

- Service providers
- Gig workers
- Platform administrators
- Local service industries

- **Benefits**

- Faster service discovery
- Better communication through voice interaction
- Fair job distribution for workers
- Increased trust and transparency in service platforms

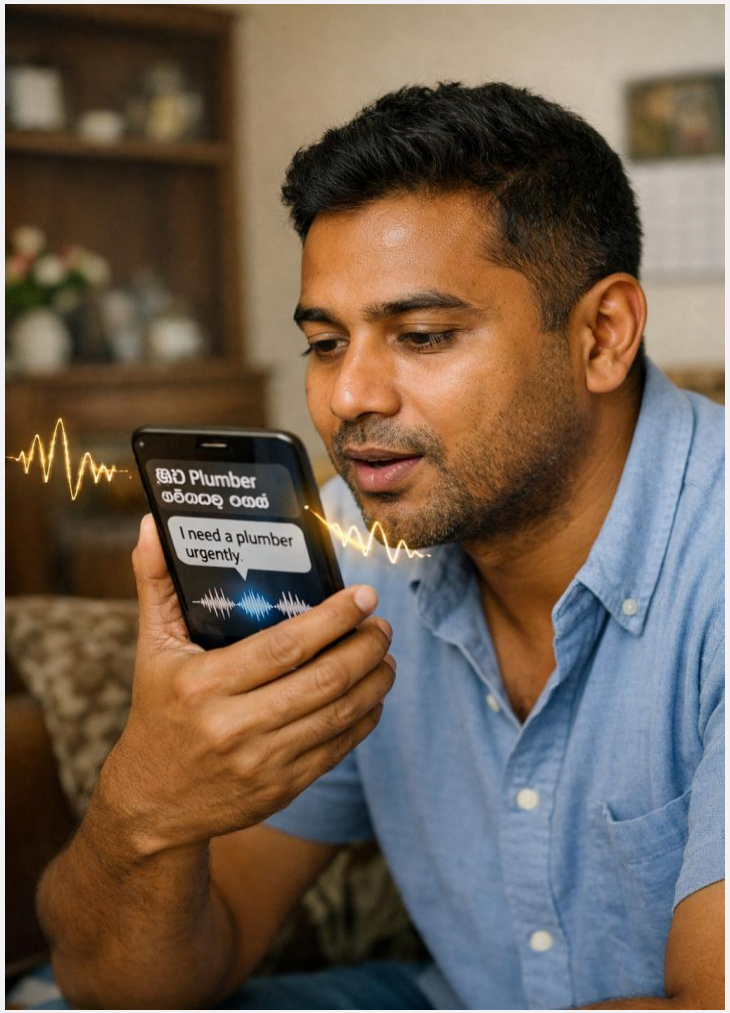
2 Minute Video Slide





Real-World Problem & Proposed Solution

- **The Problem**
- Existing service apps require typing
- Cannot understand mixed languages
- Cannot detect urgency in voice
- **Impact**
- Delayed emergency assistance
- Poor communication with service systems
- Inefficient service matching
- **Existing Solutions**
- Examples:
- Google Maps
- Uber-like service platforms
- Voice assistants
- **Limitations**
- Limited multilingual understanding
- No urgency detection
- No workforce intelligence
- **Proposed Solution**
- A **behavior-aware multilingual voice-based local service discovery platform powered by a multi-agent AI system** that supports real-world voice communication and intelligent service coordination.





Adaptive Multilingual and Urgency-Aware Voice Intent Model (Object 1)

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- Contribution to overall system:
- Enables intelligent understanding of voice requests using multilingual speech recognition and urgency detection.



Novelty + Problem Definition + Sub Objectives

Problem Definition

- Existing voice assistants fail to accurately understand real-world speech patterns in Sri Lanka where users frequently mix Sinhala and English.
- Emergency situations also involve incomplete speech such as:
- “plumber kenek fast…”
- Traditional systems cannot interpret such fragmented requests.

Novelty

- Code-switch aware voice understanding
- Incomplete speech detection
- Urgency detection using acoustic features
- Multimodal refinement using visual context

Sub Objectives

- Code-switch aware intent understanding
- Fragmented speech handling
- Urgency detection using acoustic signals
- Optional image-based context refinement

Application Domain Knowledge + Research Methodology

Application Domain
Speech Recognition
Natural Language Processing
Behaviour-Aware AI
Voice Interaction System
Research Methodology
Data collection of multilingual speech samples
Feature extraction using MFCC and acoustic analysis
Intent classification using deep learning models
Training urgency detection classifier



Technologies + Validation

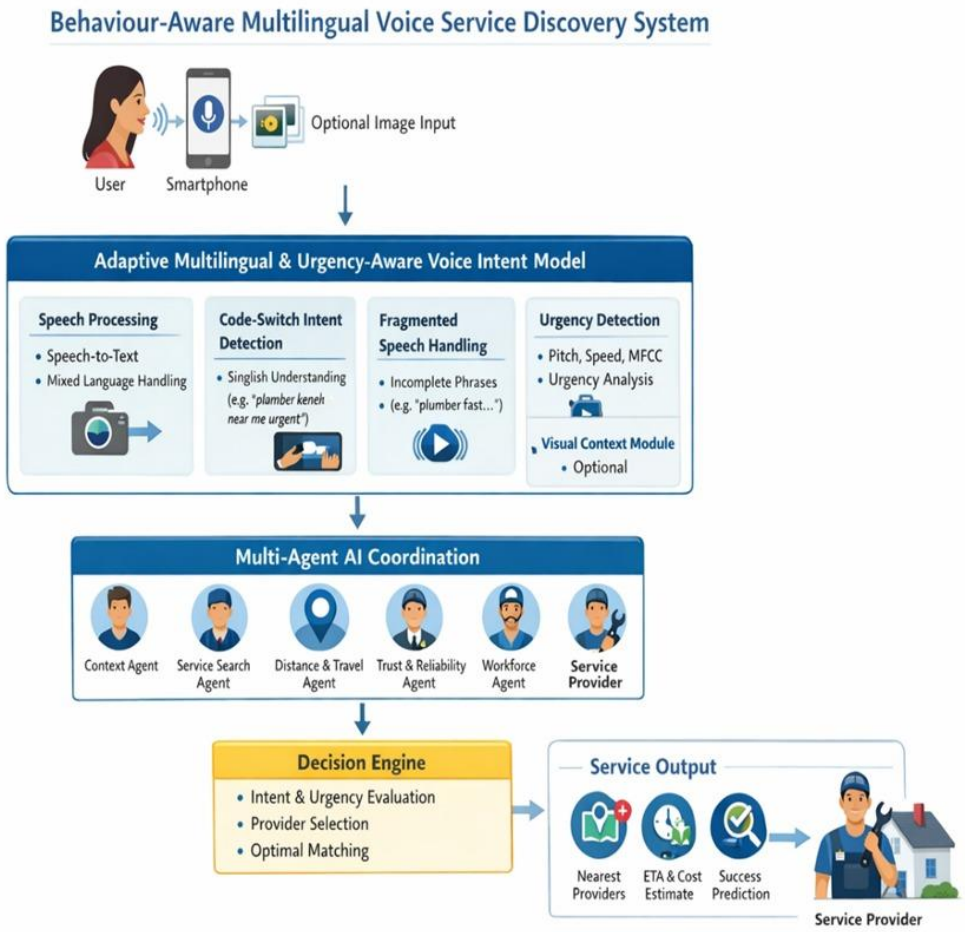
- **Technologies Used**
- Python
- Speech Recognition APIs
- Deep Learning (TensorFlow / PyTorch)
- CNN models (ResNet / EfficientNet)
- NLP libraries

- **Validation**
- Accuracy of intent classification
- Urgency detection accuracy
- User testing for real-world voice inputs

- **Data Availability**
- Multilingual speech datasets
- Custom collected Sri Lankan speech samples

- **Security & Ethics**
- Secure voice data storage
- User consent for audio data
- Ethical approval for dataset collection

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Implementation Details

System Overview

- Voice input captured from user
- Speech converted to text
- Intent detection module analyzes request
- Urgency detection model evaluates acoustic signals
- Visual context module processes optional image

Dependencies

Works with:

- Multi-Agent Coordination System
- Service Provider Database
- Interaction Layer

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Adaptive Multi-Agent Coordination Framework

Contribution to the System

- This component acts as the **decision-making engine** of the system.
- It receives the **user service request intent** from the intent detection module.
- Coordinates multiple specialized agents to analyze **context, distance, availability, and service quality**.
- Integrates outputs from agents and generates the **best service provider recommendation**.
- Improves **accuracy, responsiveness, and reliability** of service discovery.

Novelty

Adaptive Context-Aware Multi-Agent Framework for service provider recommendation.
Dynamic **context-based scoring mechanism** that adjusts ranking weights based on urgency and user context.
Predictive availability and ETA estimation using machine learning models.
Combination of **multiple decision factors** (rating, ETA, availability, reliability, response speed).
Coordinator-agent architecture enabling modular intelligent decision making.

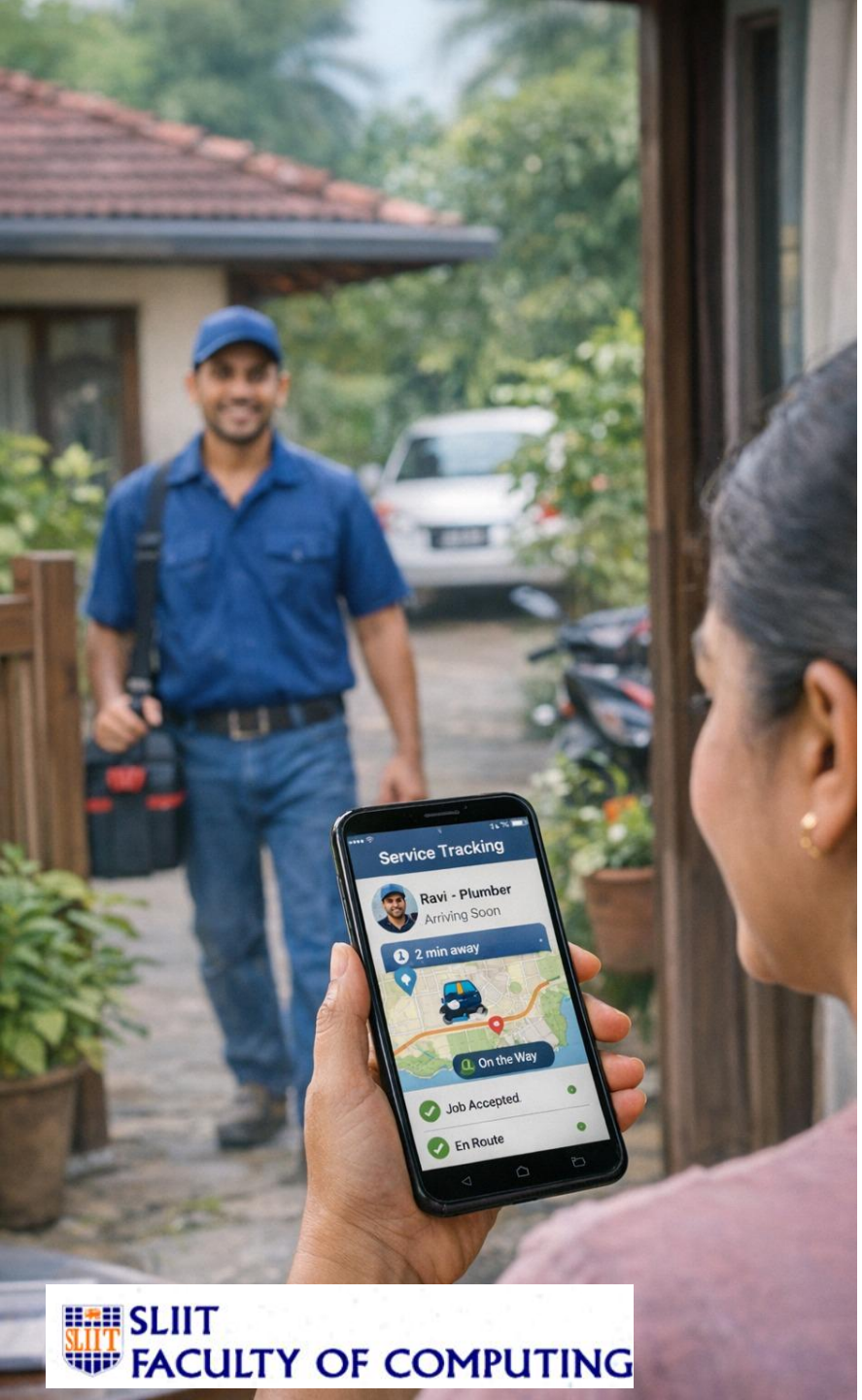
Problem Definition

- Current service discovery platforms have several limitations:
- Most systems use **static ranking algorithms**.
- Service recommendations often ignore **real-time context** such as urgency or time of day.
- Distance-based systems cannot evaluate **service quality or provider reliability**.
- Lack of intelligent coordination between **multiple decision factors**.
- Users may receive **slow, inaccurate, or unsuitable service recommendations**.

Research Objectives

Main Objective

- To design and develop an **adaptive multi-agent coordination framework** that improves the accuracy and relevance of service provider recommendations.
- **Specific Objectives**
- Design a **multi-agent architecture** for service discovery.
- Implement **context analysis** for urgency-aware recommendations.
- Develop **predictive ETA and availability models**.
- Implement an **adaptive ranking mechanism** for service provider selection.
- Evaluate system performance against **traditional static recommendation approaches**.



Research Methodology (Design Science Research)

- The research follows the **Design Science Research (DSR) methodology**.
- **Problem Identification**
  - Identify limitations in existing service discovery platforms.
- **Artifact Design**
  - Design the Adaptive Context-Aware Multi-Agent Framework.
- **Development**
  - Implement the agents, coordination mechanism, and scoring models.
- **Evaluation**
  - Evaluate system performance using experimental datasets.
- **Contribution**
  - Demonstrate improved recommendation accuracy and service reliability.

Technologies Used

- Programming & Frameworks
- Python
- FastAPI
- Multi-Agent Coordination
- CrewAI / Custom Multi-Agent Controller

Machine Learning

- Scikit-learn
- NumPy
- Pandas

Location Services

- Google Maps Distance Matrix API
- Geopy / Haversine distance
- Database
- PostgreSQL / MongoDB
- Evaluation Tools
- PyTest
- Matplotlib / Seaborn

Validation and Evaluation

- The system will be evaluated using:
- **Recommendation Quality**
- Precision
- Recall
- F1 Score
- NDCG (Ranking accuracy)
- **System Performance**
- Recommendation response time
- Service acceptance rate
- ETA prediction accuracy
- **Comparative Evaluation**
- Compare **adaptive ranking model** with **static scoring approach**.

- Availability Prediction Model
- Probability that a provider will accept the job.
- • provider historical acceptance rate
- • time of day
- • workload
- • distance to job
- • job type
- • previous response time
- XGBoost
- LightGBM • Neural ranking model

- Intelligent Ranking Model
- Replace the formula:
- Score = Rating*0.3 + Availability*0.2 + Reliability*0.2 + ResponseSpeed*0.1 + ETA*0.2

- ETA Prediction Model
- distance
- • traffic
- • time of day
- • road type
- • historical travel data

System Overview

- User submits service request
- Context agent analyzes request
- Search agent retrieves candidate providers
- Distance agent calculates travel time
- Availability agent checks provider status
- Ranking agent selects best provider
- Provider receives service request

Dependencies

This component interacts with:

- Voice Intent Model (Objective 1)
- Workforce Intelligence System (Objective 3)
- Interaction Layer (Objective 4)





AI-Driven Employment Empowerment & Workforce Intelligence Framework Object (3)

- **Gamage D G P U**
- IT 22245588

- **Contribution**
- This component enhances the platform by introducing workforce intelligence mechanisms that promote fair job distribution, employment sustainability, and provider performance monitoring.



Novelty, Problem Definition & Sub Objectives

Problem Definition

- Many gig-based service platforms concentrate job opportunities among a small group of providers while others receive very few tasks.
- This results in **income instability and unfair job distribution.**

Novelty

- Employee identity verification
- Fairness-aware job allocation
- Dynamic provider reputation system
- Workforce demand prediction
- Income stability analysis

Sub Objectives

- Intelligent employee bank creation
- Fairness-aware task allocation
- Dynamic reputation evolution model
- Workforce demand prediction
- Income stability prediction



Application Domain & Research Methodology

Application Domain

- Workforce Intelligence Systems
- Predictive Analytics
- Gig Economy Platforms
- Machine Learning Optimization

Research Methodology

- The system analyzes historical service data to understand workforce behaviour.
- Machine learning models are used to predict demand patterns and optimize task allocation.
- Fairness algorithms ensure equal opportunity distribution among providers.

Technologies, Validation, Data & Ethics

Technologies Used
- Python
- Machine Learning models
- Predictive analytics tools
- Database analytics systems

Validation

Evaluation is conducted using:
- Job distribution fairness metrics
- Worker income stability
- Provider satisfaction

Data Availability
- Historical service records
- Worker performance data
- Demand patterns

Security & Ethical Considerations
- Protection of worker data
- Fair algorithmic decision making
- Transparent reputation scoring

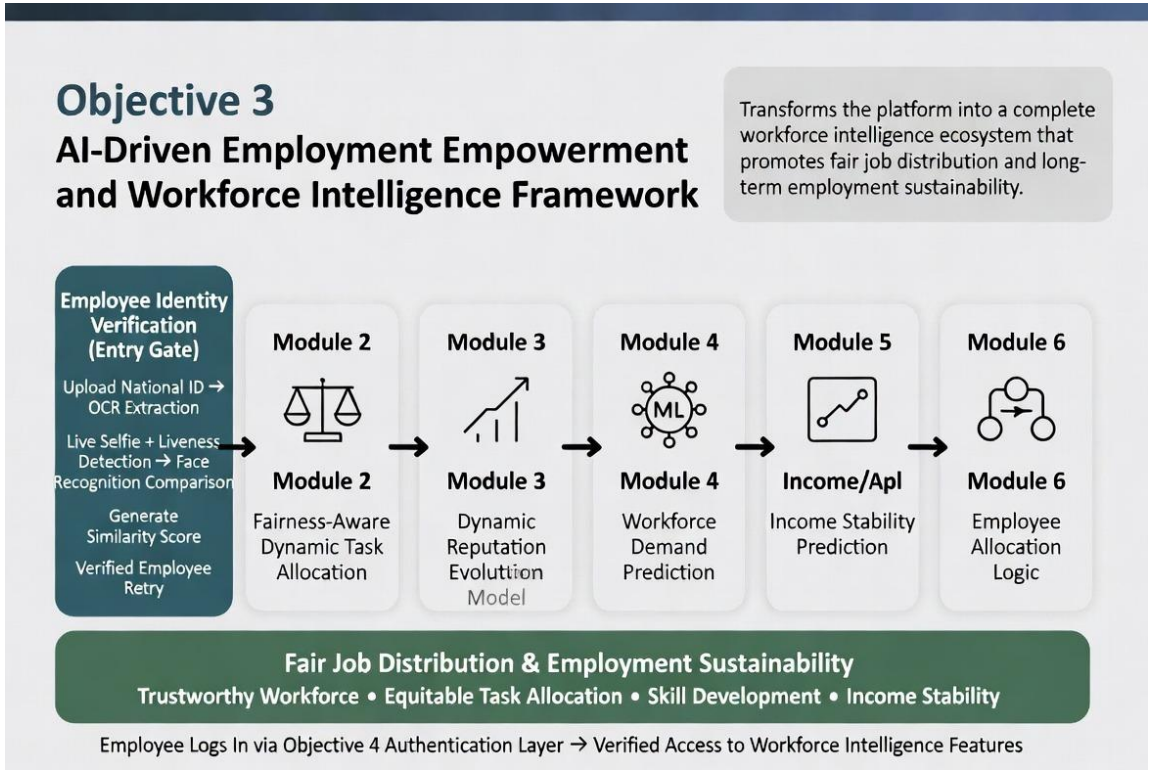


System Overview

- Worker profiles stored in employee bank
- System analyzes service history
- Fairness model distributes tasks
- Reputation scores updated dynamically
- Demand prediction identifies workforce needs

Dependencies

- • Multi-Agent Coordination System
  • Service Request Data
  • Interaction Layer



Behavior-Aware Intelligent Interaction Layer



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Contribution

- This component improves user experience and transparency by providing intelligent dashboards, service guidance, security mechanisms, and predictive insights.



Contribution to the System

This component enhances the system by introducing intelligent decision support and personalized interaction for users.

Provides **smart pre-service guidance** that helps users resolve minor issues before requesting professional service.

Uses **machine learning–based severity detection** to classify issues as Critical, Moderate, or Low.

Predicts the **likelihood of successful service completion** based on provider reliability, distance, and past performance.

Generates **estimated service costs** to improve financial transparency and help users make informed decisions.

Delivers a **personalized dashboard and secure user environment**, improving user experience, trust, and system reliability.

Novelty, Problem Definition & Sub Objectives

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Problem Definition

- Many service platforms lack transparency and intelligent interaction features. Users often struggle to understand service recommendations, estimated costs, and service success probability.

Novelty

- Behaviour-aware user dashboards
- Dynamic service success prediction
- Smart service guidance system
- Suspicious login detection
- Profile-based service cost estimation

Sub Objectives

- User and employee authentication system
- Intelligent user dashboard
- Service success prediction model
- Smart service guidance module
- Account protection and security monitoring



Novelty

- **Smart Service Guidance Module** providing intelligent pre-service assistance to help users resolve minor issues before requesting technicians
- **Machine Learning–based Severity Detection** that classifies issues into Critical, Moderate, and Low levels using contextual problem features.
- **Predictive Success Estimation** that analyzes provider reliability, travel distance, and historical performance to estimate service completion probability.
- **AI-driven Cost Estimation Model** that predicts approximate service costs based on service type, urgency, location, and provider history.
- **Behaviour-Aware Personalized Interaction Layer** that adapts dashboards, recommendations, and system responses based on user behavior patterns.

Technologies, Validation, Data & Ethics



Technologies Used

- • Web and mobile application frameworks
  • Machine learning prediction models
  • Authentication systems
  • Database analytics

Validation

- Evaluation metrics include:
- • User satisfaction
  • Prediction accuracy
  • System usability

Data Availability

- • User interaction logs
  • Service request history
  • Provider performance records

Security & Ethical Considerations

- • Secure authentication
  • User privacy protection
  • Ethical handling of personal data

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Application Domain & Research Methodology

Application Domain

- Human-Computer Interaction
- Behaviour-Aware Systems
- Predictive Analytics
- Security Systems

Research Methodology

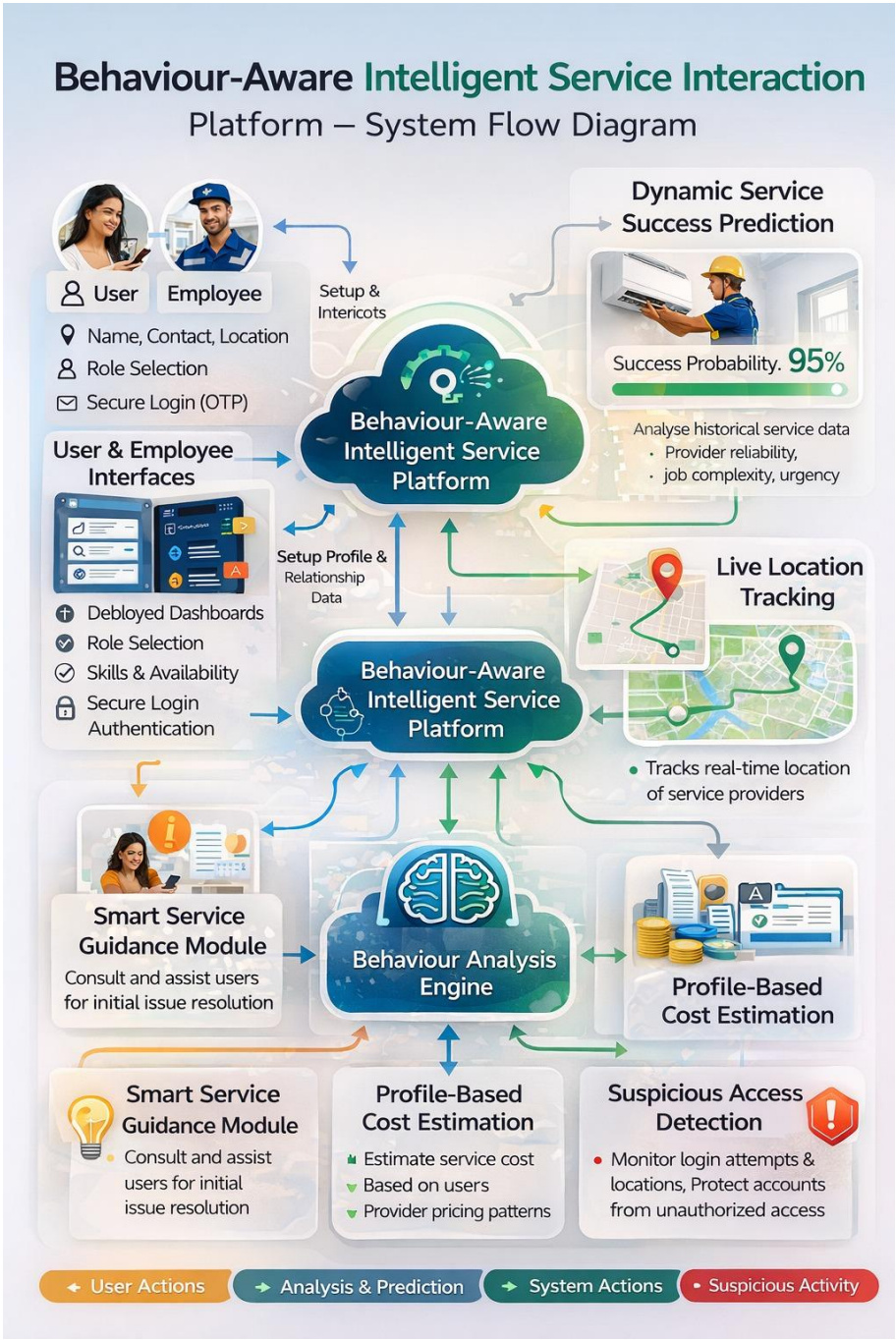
- User behaviour patterns and service data are analyzed to provide personalized recommendations and predictions.
- Security algorithms monitor system access patterns to detect suspicious behaviour.

System Overview

User logs into platform
Dashboard displays service insights
System predicts service success probability
Guidance module provides recommendations
Security module monitors login behavior
Cost estimation module predicts service price

Dependencies

- Voice Intent Model
- Multi-Agent Coordination System
- Workforce Intelligence Framework



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Commercialization

Target Market

- Urban residents
- Emergency service seekers
- Home maintenance users
- Gig service workers

Customer Profile

- Smartphone users
- Individuals requiring urgent local services

Value Proposition

- Fast voice-based service discovery
- Multilingual communication
- Intelligent service provider selection

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- * Landsberger, J. Detecting Urgency in Speech with Personalized Acoustic Features, SEMDIAL 2020. https://www.semdial.org/anthology/Z20-Landesberger_semdial_0027.pdf ([semdial.org][6])
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Gantt Chart

Behaviour-Aware Urgency-Sensitive Multilingual Voice-Based Local Service Discovery and Workforce Empowerment System Using a Multi-Agent AI Framework

Tasks	Oct. 2025	Nov. 2025	Dec. 2025	Jan. 2026	Feb 2026	Mar 2026	Aug 2026	Oct. 2026	Dec. 2026	Dec 2026
Research Project Workshop										
Select a Topic										
Find Supervisor & Co-supervisor										
Group Register										
Gathering Information										
Topic Submission										
Project Charter Form Submission										
Proposal Submission										
Check List Submission										
Progress Presentation 1										
Final Checklist Submission										
Progress Presentation 1										
Progress Presentation										
Final Report Submission										
Website Submission										
Final Presentation & Viva										
Research Paper Submission										
Paper Publication Evidence										
Research Paper Submission										

Research Development Integration Testing Documentation



Thank you !